CLINICAL · NECK, THYROID & SALIVARY

Congenital neck masses

Most neck lumps in children are reactive nodes. The congenital masses that remain are explained by embryology. A midline cyst that rises when the tongue is put out is a thyroglossal cyst. A smooth lump in front of the upper third of sternomastoid is a branchial cyst, and a pit along its lower anterior border is a branchial sinus. A soft, transilluminant, multilocular swelling in a baby's posterior triangle is a lymphatic malformation. Each is cured by removing the whole tract, and each recurs if the tract is left behind.

AT A GLANCE

- Midline moving with the tongue: thyroglossal.
- Lateral: branchial.
- Treat infection first, then excise.
- Sistrunk removes the central hyoid.

1 CHAPTER ONE Embryology

Arches, clefts and pouches.

1.1 The branchial apparatus

Between the fourth and eighth weeks, five (some count six) *branchial arches* form on each side of the primitive pharynx. They are separated on the outside by **clefts** lined with ectoderm and on the inside by **pouches** lined with endoderm, and each arch contains a cartilage bar, an artery, a nerve and muscle-forming mesoderm. The second arch then overgrows the third and fourth to form the smooth contour of the neck, temporarily enclosing a space, the *cervical sinus of His*.

When a cleft, a pouch or the cervical sinus fails to obliterate, it leaves a remnant: a *cyst* with no opening, a *sinus* with one opening, inside or out, or a *fistula* with both. The skin openings always lie *anterior to the sternomastoid*, and the tract runs deep to platysma.

ARCHES, THEIR NERVES AND DERIVATIVES

ARCH	NERVE	SKELETAL AND MUSCLE DERIVATIVES	ARTERY
First (mandibular)	Trigeminal (V3)	Malleus head and neck, incus body and short process, mandible, sphenomandibular ligament; muscles of mastication, mylohyoid, anterior digastric, tensor tympani, tensor palati	Degenerates (part of maxillary)
Second (hyoid)	Facial (VII)	Manubrium of malleus, long process of incus, stapes superstructure, styloid process, stylohyoid ligament, lesser cornu and upper body of hyoid; facial muscles, stapedius, stylohyoid, posterior digastric	Stapedial (degenerates)
Third	Glossopharyngeal (IX)	Greater cornu and lower body of hyoid; stylopharyngeus	Common and internal carotid
Fourth	Superior laryngeal (X)	Thyroid and cuneiform cartilages; cricothyroid, pharyngeal constrictors, cricopharyngeus	Aortic arch (left), proximal subclavian (right)
Sixth	Recurrent laryngeal (X)	Cricoid, arytenoid and corniculate cartilages; intrinsic laryngeal muscles	Pulmonary arteries, ductus arteriosus

POUCH DERIVATIVES

POUCH	DERIVATIVES
First	Eustachian tube, middle ear, mastoid air cells
Second	Palatine tonsil and tonsillar fossa
Third	Inferior parathyroids (dorsal wing), thymus (ventral wing)
Fourth	Superior parathyroids; ultimobranchial body with the C cells

WHY THE INFERIOR PARATHYROID IS "3" AND LIES BELOW "4"

The third pouch gives rise to both the inferior parathyroid and the thymus. As the thymus descends into the chest, it drags the third-pouch parathyroid below the fourth-pouch one. That long journey explains why inferior glands are so often ectopic.

1.2 The thyroid and tongue

The thyroid begins at the *foramen caecum* at about four weeks and descends through the tongue base, passing in front of, through or behind the developing hyoid. The *thyroglossal duct* that connects it to the tongue normally disappears by the sixth to eighth week. The anterior two-thirds of the tongue come from the first-arch lingual swellings and the tuberculum impar, the posterior third from the copula of the second to fourth arches, and the muscles from the occipital somites, which is why they are supplied by the hypoglossal nerve.

2 CHAPTER TWO Midline masses

Thyroglossal and dermoid.

2.1 Thyroglossal duct cyst

The thyroglossal cyst is the commonest congenital neck mass. It usually presents in childhood but can appear at any age, as a smooth, round, cystic *midline* swelling of 2 to 4 cm, usually *at or just below the hyoid* and sometimes slightly to the left. Because it is tethered to the foramen caecum, it *rises on swallowing and on protruding the tongue*. It may become infected after an upper respiratory infection, and if it ruptures or is incised it leaves a thyroglossal sinus or "fistula". It can lie anywhere along the tract: lingual, suprahyoid, at the hyoid, subhyoid (the commonest) or suprasternal.

Two rarities matter. The cyst may contain the *only functioning thyroid tissue*, and about 1% contain carcinoma, almost always papillary. Ultrasound therefore confirms not only the cyst but also a *normal thyroid gland*, with thyroid function tests; scintigraphy is done if the gland is not seen, and solid areas in adults are sampled by FNA.

The treatment is the **Sistrunk operation**: excision of the cyst and tract *together with the central body of the hyoid bone* and a core of tongue-base muscle along the tract up to the foramen caecum. Simple cystectomy recurs in up to half of cases; the Sistrunk operation brings recurrence below 5 to 10%. Infection is first settled with aspiration or antibiotics, and the operation done afterwards.

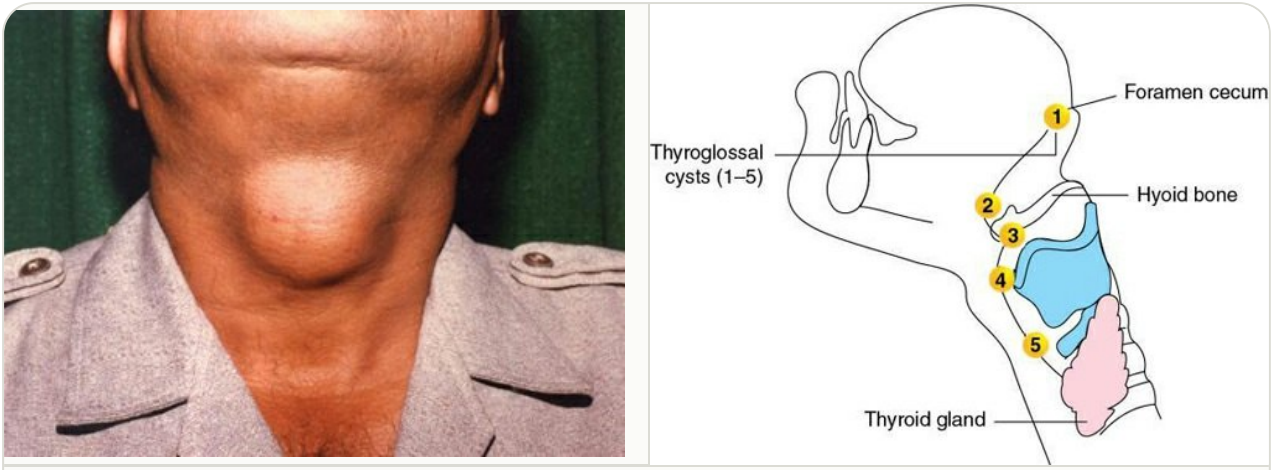
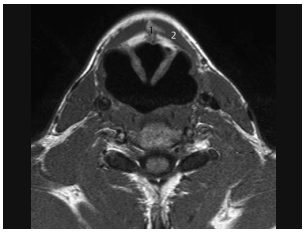


FIGURE 1 First, a thyroglossal cyst in the midline of the neck. Second, sites of thyroglossal cysts along the tract (subhyoid is the commonest).
Dhingra, figs 77.3 and 77.4

IMAGING IN THE APP 1 case

Thyroglossal duct cyst on imaging.



Thyroglossal Duct Cyst
Congenital · 11 images

OPERATIVE GUIDES IN THEATRE 1 operation

The Sistrunk operation.

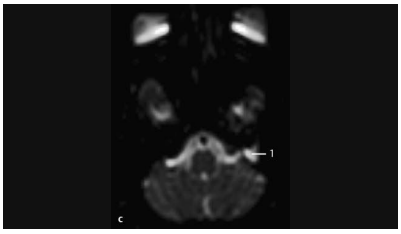
Thyroglossal Duct Cyst
Sistrunk

2.2 Dermoid cyst

A dermoid cyst is lined by skin with its adnexa, formed from ectoderm trapped at a line of fusion. It lies in the submental region or the floor of mouth (the *sublingual dermoid*), or at the suprasternal notch, and it does not move with tongue protrusion. A sublingual dermoid above the mylohyoid can push the tongue up and must be told apart from a ranula. It is excised through the mouth or the neck according to its relation to the mylohyoid.

IMAGING IN THE APP 1 case

Epidermoid and dermoid cysts.



(Epi)dermoid Cyst

Infection & inflammation · 10 images

2.3 Other midline masses

A **teratoma** in a neonate can be huge and obstruct the airway, and when it is found antenatally an EXIT (ex-utero intrapartum treatment) delivery is planned. Ectopic thyroid, a Delphian node, a bronchogenic cyst at the suprasternal notch, and a thymic cyst, usually lateral and deep to the lower third of the sternomastoid, complete the differential.

3 CHAPTER THREE Branchial anomalies

Lateral cysts, sinuses and fistulae.

3.1 Second cleft anomalies

About 90% of branchial anomalies arise from the second cleft. The **cyst** usually appears in older children and young adults as a smooth, fluctuant, non-tender swelling at the junction of the upper and middle thirds of the anterior border of the sternomastoid. It often enlarges with upper respiratory infections and may become infected, and its fluid contains cholesterol crystals.

The **sinus or fistula** is seen from infancy as a pit on the *lower third of the anterior border of the sternomastoid* that discharges mucus. Its tract runs up along the carotid sheath deep to platysma, passes *deep to the second-arch derivatives* (the external carotid, stylohyoid and posterior digastric) and *superficial to the third-arch derivatives* (the internal carotid and glossopharyngeal nerve), so that it runs between the two carotids, crosses over the hypoglossal nerve, and ends in the *tonsillar fossa*.

The anomaly is excised completely once infection has settled. A fistula is removed through an elliptical incision around the opening, with a second, higher *step-ladder* incision to follow the tract. Recurrence follows incomplete excision or surgery during infection.



FIGURE 2 A left branchial sinus discharging pus.
Dhingra, fig. 77.5

IMAGING IN THE APP 1 case

Branchial cleft cyst on imaging.



Branchial Cleft Cyst

Congenital · 12 images

THE ADULT "BRANCHIAL CYST"

A new cystic lateral neck mass in an adult over about 40 is far more often a *cystic metastasis* from HPV-related oropharyngeal carcinoma or papillary thyroid carcinoma than a branchial cyst. Examine the tonsils and tongue base, aspirate under ultrasound, test p16 and thyroglobulin, and image before excision.

3.2 First cleft anomalies

First cleft anomalies are duplications of the external ear canal. **Work type I** is ectodermal, a duplicate canal running parallel to the external canal; **Work type II** is ectodermal and mesodermal and contains cartilage. They present as a cyst, sinus or fistula near the angle of the mandible or within the parotid, draining into or near the ear canal, often as recurrent abscesses in the parotid region or as otorrhoea. Because the tract runs close to, and sometimes deep to, the facial nerve, excision needs a parotidectomy approach with the nerve identified first.

3.3 Third and fourth pouch anomalies

Third and fourth pouch anomalies open into the *pyriform sinus*, almost always on the left. They present as recurrent *left-sided neck abscesses*, as *acute suppurative thyroiditis* in children, or as cysts low in the lateral neck. A third-arch tract runs deep to the internal carotid and glossopharyngeal nerve and superficial to the hypoglossal nerve, piercing the thyrohyoid membrane above the internal laryngeal nerve. A

fourth-arch tract passes below the superior laryngeal nerve and loops round the subclavian artery on the right or the aortic arch on the left.

The diagnosis is made with a barium swallow, CT and direct laryngoscopy to see the pyriform opening. The sinus opening can be closed endoscopically by chemical or electrocautery, or the tract excised openly with a partial thyroidectomy.

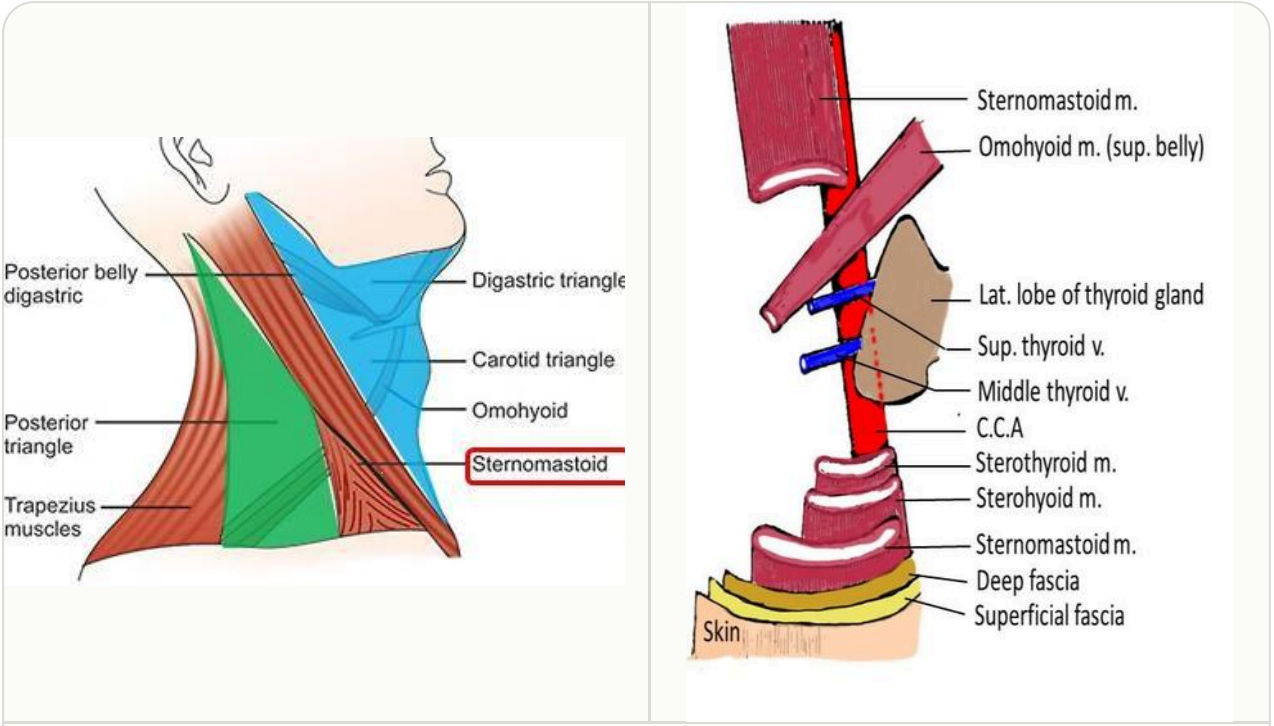


FIGURE 3 The triangles of the neck, and the common carotid artery and its relations, which guide the course of branchial tracts.

Anatomy atlas · ENT Steps p. 270, Doss p. 136

OPERATIVE GUIDES IN THEATRE 1 operation

Branchial cleft cysts and sinuses.

Branchial Cleft Cysts and Sinuses

Branchial surgery

BRANCHIAL ANOMALIES COMPARED

TYPE	EXTERNAL OPENING	INTERNAL OPENING	KEY RELATIONS
First cleft	Angle of mandible, near the ear canal	External auditory canal	Close to the facial nerve
Second cleft (commonest)	Lower third, anterior border of sternomastoid	Tonsillar fossa	Between the external and internal carotids, over XII

TYPE	EXTERNAL OPENING	INTERNAL OPENING	KEY RELATIONS
Third pouch	Lower anterior border of sternomastoid	Upper pyriform sinus	Behind the internal carotid, over XII, above the internal laryngeal nerve
Fourth pouch	Low in the neck (rare)	Apex of the pyriform sinus	Below the superior laryngeal nerve, around the subclavian or aorta

4 CHAPTER FOUR Lymphatic and vascular

Malformations and haemangiomas.

4.1 Lymphatic malformation (cystic hygroma)

A lymphatic malformation arises from sequestration of the primitive jugular lymph sacs. About half are present at birth and 90% appear by two years. They occur most often in the *posterior triangle* and extend into the floor of mouth and tongue, the parotid, the axilla and the mediastinum.

Macrocystic lesions, the classic cystic hygroma, are soft, multilocular, compressible and *brilliantly transilluminant*. **Microcystic** lesions infiltrate the tongue and face, and the overlying mucosa shows clear or blood-stained *vesicles*. Both swell suddenly with infection or bleeding into the cysts, and large ones can obstruct labour, the airway or feeding. MRI shows the extent, and when antenatal ultrasound detects a large lesion an EXIT delivery is planned.

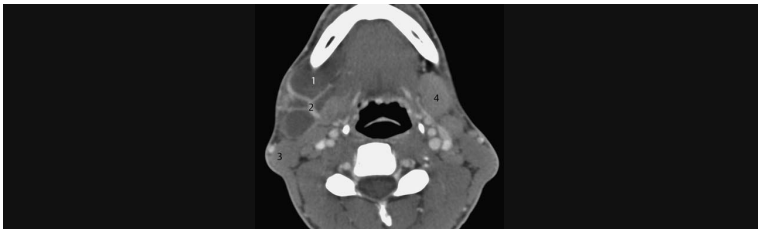
Macrocystic disease responds to *sclerotherapy* with bleomycin, doxycycline, OK-432 or ethanol. *Surgical excision* preserves nerves and vessels; recurrence is low after complete excision and about half after partial excision, and several stages may be needed. Extensive microcystic disease is treated with *sirolimus*, mucosal vesicles with laser, and airway obstruction with a tracheostomy.



FIGURE 4 A cystic hygroma in an infant.
Dhingra, fig. 77.7

IMAGING IN THE APP 1 case

Cystic hygroma (lymphatic malformation).



Lymphatic Malformation (Cystic Hygroma, Lymphangioma)
Congenital · 14 images

4.2 Infantile haemangioma and other vascular anomalies

The infantile haemangioma, the commonest tumour of infancy, appears after birth, proliferates for about a year and then slowly involutes. A lesion in a *beard distribution* means a subglottic haemangioma until proved otherwise, and a large segmental facial lesion suggests *PHACES syndrome*. Propranolol is the treatment. Venous

malformations, capillary malformations and AVMs, which grow with the child, are covered with the vascular anomalies.

5 CHAPTER FIVE Other paediatric masses

Nodes, glands, muscle.

5.1 The sternomastoid tumour and the thymus

Fibromatosis colli, the sternomastoid tumour, is a firm spindle-shaped mass within the sternomastoid of a baby two to four weeks old, often after a difficult delivery. It causes *torticollis*, with the head tilted towards the lesion and the chin turned away, and ultrasound confirms it. Stretching physiotherapy resolves most cases, and persistent cases need tenotomy to prevent plagiocephaly and facial asymmetry. A **thymic cyst** is a cervical remnant lying along the line of thymic descent from the angle of the jaw to the mediastinum, usually on the left, and it may extend into the chest.

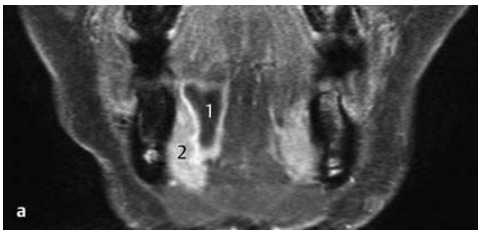
5.2 Nodes and malignancy

Reactive and infective nodes are the commonest childhood neck masses: suppurative adenitis, atypical mycobacteria and cat-scratch disease. **Kawasaki disease**, with more than five days of fever, conjunctivitis, mucosal changes, a rash and a large node, must not be missed, because coronary aneurysms follow when it goes untreated.

Malignancy is uncommon but important: lymphoma, especially Hodgkin lymphoma in adolescents, rhabdomyosarcoma, neuroblastoma, thyroid carcinoma, and nasopharyngeal carcinoma in endemic areas. A hard, fixed or matted node over 2 to 3 cm, a supraclavicular node, growth over two to six weeks despite antibiotics, and B symptoms are the warnings that call for biopsy. Salivary masses in children include parotid haemangioma, ranula and juvenile recurrent parotitis.

IMAGING IN THE APP 1 case

Ranula of the sublingual gland.



Ranula of the Sublingual Gland

Infection & inflammation · 10 images

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Embryology:** the arches with their nerves and derivatives; the pouches and what they form.
- **Thyroglossal cyst:** clinical signs; check for a normal thyroid; the Sistrunk operation and why the hyoid is removed.
- **Second branchial fistula:** its course and the step-ladder excision.
- **Third and fourth pouch sinuses:** recurrent left neck abscess and thyroiditis.
- **Cystic hygroma:** features and management.
- **The adult "branchial cyst":** a metastasis until proved otherwise.

6.2 Common pitfalls

COMMON PITFALLS

- Simple excision of a thyroglossal cyst without the hyoid and tract.
- Removing a thyroglossal "cyst" that is the patient's only thyroid tissue.
- Incising and draining a branchial cyst or atypical mycobacterial node.

- Missing a first cleft anomaly as a recurrent "parotid abscess" or discharging ear.

Written for this app from Oxford Textbook of Otolaryngology ch. 81, Oxford Specialist Handbook ch. 23, Oxford Handbook of ENT ch. 17, Dhingra ch. 77 and K. J. Lee ch. 40, 51. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

NECK: TRIANGLES, FASCIA & NODES · CONGENITAL

Branchial Cleft Cyst

IMAGING FINDINGS

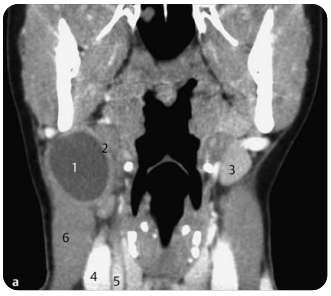
- Cyst at the angle of the mandible in front of the sternocleidomastoid
- Lateral to the carotid sheath, behind the submandibular gland
- Thick wall when infected

TOP DIFFERENTIALS

- Cystic nodal metastasis in an adult
- Lymphatic malformation
- Thymic cyst

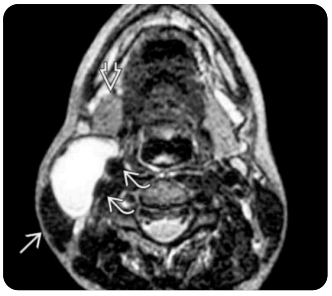
CHECKLIST

- In adults: exclude HPV oropharyngeal cancer
- Tract towards the tonsillar fossa



CT with intravenous contrast enhancement, coronal (a) and axial (b).

Dunnebier, Imaging for…



Axial T2WI MR shows a well-defined hyperintense mass anterior to the right SCM, posterolateral to the right submandibular gland, and…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 602.

THYROID & PARATHYROID · CONGENITAL

Thyroglossal Duct Cyst

IMAGING FINDINGS

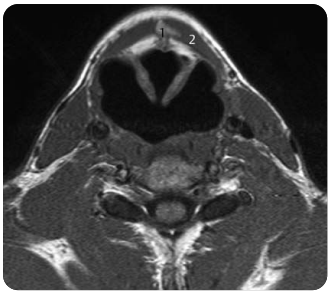
- Midline or paramedian cyst at or below the hyoid
- Infrahyoid cysts embedded in the strap muscles
- Thick wall or enhancement if infected

TOP DIFFERENTIALS

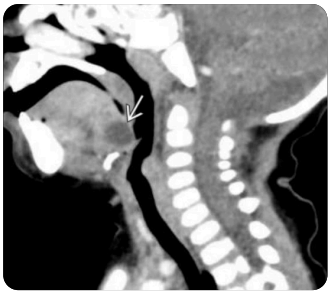
- Dermoid or epidermoid
- Lymphatic malformation
- Lingual thyroid

CHECKLIST

- Normal thyroid in the lower neck on ultrasound
- Solid nodule in the wall in an adult: carcinoma



MRI, T1-weighted, axial. Small lesion (1) with high signal intensity on T1-weighted images, probably due to presence of protein-rich fluid, situate…



Sagittal CECT shows a well-defined cysticappearing mass at the midline base of the tongue. This was incidentally found on a CT performed…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 590.

NECK: TRIANGLES, FASCIA & NODES · CONGENITAL

Lymphatic Malformation (Cystic Hygroma, Lymphangioma)

IMAGING FINDINGS

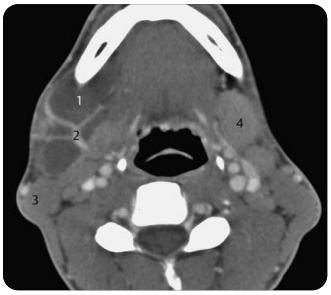
- Multicystic, septated, transspatial mass
- Fluid–fluid levels after bleeding
- Insinuates between structures without mass effect

TOP DIFFERENTIALS

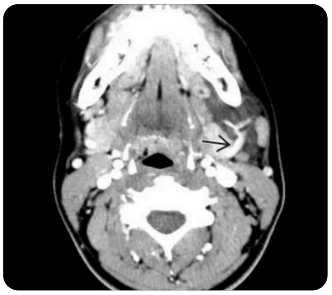
- Infantile haemangioma
- Venous malformation (phleboliths)
- Abscess

CHECKLIST

- Airway and mediastinal extent
- Macrocystic versus microcystic (sclerotherapy)



CT with intravenous contrast enhancement, axial. There is hypodense multilobular cystic structure (1) at the level of the…



Axial CECT in a 22-yearold woman with an angle of a mandible mass reveals a low-density submandibular space

Koch et al., Diagnostic Imaging: Head…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 580.

INNER EAR & IAC · INFECTION & INFLAMMATION

(Epi)dermoid Cyst

IMAGING FINDINGS

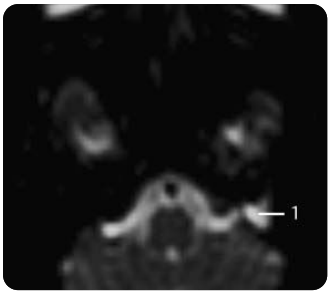
- CSF-like lesion that insinuates around nerves and vessels
- Incomplete suppression on FLAIR
- Bright on diffusion; no enhancement

TOP DIFFERENTIALS

- Arachnoid cyst (follows CSF on every sequence, DWI dark)
- Cystic schwannoma

CHECKLIST

- Encasement of cranial nerves and vessels
- Extent across cisterns before surgery



Patient with progressive and asymmetrical perceptive hearing loss.
Dunnebier, Imaging for Otolaryngologists, Fig. 5.9 a



Axial CECT shows a large CPA epidermoid cyst. Note that this non-enhancing low-density lesion appears to invade the left cerebellar…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1202.